Do Protein Language Models Implicitly Learn Binding Thermodynamics? A Probing Study

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Abstract

Protein language models (PLMs) such as ESM-2 and ProtT5, trained exclusively on amino acid sequences, develop representations that capture structural and functional properties far beyond what their training objective explicitly supervises. Whether this implicit learning extends to thermodynamic properties – specifically binding affinity – remains an open and fundamental question. Binding affinity is a continuous thermodynamic quantity defined over a specific protein-ligand pair, never observed during PLM pretraining, making it categorically distinct from properties like binding site identification that PLMs are known to encode. We directly interrogate this by pairing frozen PLM protein embeddings with a fixed ligand representation and applying linear probes to predict affinity, deliberately avoiding task-specific fine-tuning. Through layer-wise and residue-level analysis we characterize how much thermodynamic signal exists within frozen PLM representations and where it localizes — providing a principled interpretability foundation that existing black-box affinity prediction methods have not established.

1. Introduction

Protein language models (PLMs) such as ESM-2 and ProtT5, trained on hundreds of millions of sequences through self-supervised objectives, develop representations that generalize well beyond their training objective — capturing secondary structure, localization signals, enzyme function, and binding site residues (Garcia & Ansuini, 2025; Simon & Zou, 2024) without explicit supervision. This raises a fundamental question: does this implicit learning extend to binding affinity?

Binding affinity is categorically harder than the properties PLMs are known to encode. It is a continuous thermodynamic quantity defined over a specific protein-ligand pair, never observed during PLM pretraining. Unlike binding site identification — a structural property intrinsic to the protein sequence — affinity requires reasoning about a ligand the PLM has never seen. Yet through co-evolutionary statistics, PLMs may implicitly encode physicochemical properties of binding pockets that are partially predictive of binding strength. Wang et al. (Wang et al., 2024) showed partial separability between binding and non-binding residues in frozen ESM-2 embeddings, though only for binary site classification with no affinity quantification.

Existing affinity prediction methods — ConPLex (Singh et al., 2023), BAPULM (Meda & Farimani, 2024), CAPLA (Jin et al., 2023) — leverage PLM embeddings but apply task-specific training on top, making it impossible to isolate what the PLM itself contributes versus what the downstream model additionally learns.

We directly address this by pairing frozen PLM protein embeddings with a fixed ligand representation and applying linear probes to predict affinity, with no fine-tuning. Critically, we compare against a naive protein representation baseline, amino acid composition and physicochemical descriptors, holding the ligand representation fixed. The performance delta between PLM embeddings and this baseline isolates the PLM's specific contribution. Through layer-wise and residue-level analysis we further localize where within the PLM this signal concentrates — directly informing the design of computationally efficient affinity prediction pipelines that extract only the representations where binding information lives. A positive result establishes that sequence co-evolution implicitly encodes binding-relevant information. A negative result is equally informative — establishing that affinity is irreducible to sequence statistics alone and requires explicit supervision or structural information. Either outcome provides an interpretability foundation that performance-oriented methods have not established.

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